

**MINISTRY OF EDUCATION AND TRAINING
FPT UNIVERSITY**

**A TWO-TIER COGNITIVE ARCHITECTURE FOR
AUTOMATED THREAT DETECTION AND RESPONSE
USING AGENTIC AI**

by

Nguyen Duc Binh

A thesis submitted in conformity with the requirements
for the degree of Master of Software Engineering

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Supervisor:
Dr. Bui Van Hieu
Dr. Dang Van Hieu

Abstract

In the contemporary cybersecurity landscape, Security Operations Centers (SOCs) are consistently overwhelmed by the sheer volume of network alerts, leading to severe alert fatigue and the potential oversight of critical Advanced Persistent Threats (APTs). To address the inherent limitations of traditional, static rule-based systems and the latency bottlenecks of processing raw logs directly through Large Language Models (LLMs), this thesis introduces SENTINEL: A Two-Tier Cognitive Architecture for Automated Threat Detection and Response. The proposed framework operates by deploying Welford’s online statistical algorithm at the Transport Tier (Tier-1) to filter benign traffic at wire speed with minimal computational overhead. Subsequently, the Cognitive Agent Tier (Tier-2) leverages a LangGraph-driven AI Agent equipped with a Dual Retrieval-Augmented Generation (Dual-RAG) mechanism—integrating MITRE ATT&CK and NIST guidelines—to perform deep, contextual threat reasoning and mitigation.

Evaluated on the CSE-CIC-IDS2018 and DAPT2020 datasets, SENTINEL demonstrates substantial improvements in both detection accuracy and processing efficiency. Furthermore, the system incorporates robust AI security guardrails, including Delimited Data Encapsulation and HMAC log chaining, achieving highly effective mitigation against known adversarial AI manipulation attacks and ensuring forensic data integrity. The findings underscore the efficacy of combining deterministic statistical filters with explainable Agentic AI, offering a highly scalable and secure automation solution for modern network defense.

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List of Abbreviations

Abbreviation	Full Description
AI	Artificial Intelligence
APT	Advanced Persistent Threat
API	Application Programming Interface
C2	Command and Control
CIDR	Classless Inter-Domain Routing
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CSV	Comma-Separated Values
CTI	Cyber Threat Intelligence
CVE	Common Vulnerabilities and Exposures
DAG	Directed Acyclic Graph
DAPT	Dynamic/Delayed Advanced Persistent Threat
DDoS	Distributed Denial of Service
DL	Deep Learning
DoS	Denial of Service
EDR	Endpoint Detection and Response
FAISS	Facebook AI Similarity Search
FSM	Finite State Machine
GenAI	Generative Artificial Intelligence
GGUF	GPT-Generated Unified Format
GPU	Graphics Processing Unit
HITL	Human-in-the-Loop
HMAC	Hash-based Message Authentication Code
HTML	HyperText Markup Language
IDS	Intrusion Detection System
IOC	Indicator of Compromise
IP	Internet Protocol
IPS	Intrusion Prevention System
JSON	JavaScript Object Notation
LLM	Large Language Model
LRU	Least Recently Used
LSTM	Long Short-Term Memory
MAS	Multi-Agent System
ML	Machine Learning
NGFW	Next-Generation Firewall
NIST	National Institute of Standards and Technology
OWASP	Open Web Application Security Project
PCAP	Packet Capture
RAG	Retrieval-Augmented Generation
RAGAS	Retrieval Augmented Generation Assessment
RAM	Random Access Memory
RBA	Runbook Automation
RCE	Remote Code Execution

RRF	Reciprocal Rank Fusion
SCA	Software Composition Analysis
SHA	Secure Hash Algorithm
SIEM	Security Information and Event Management
SOC	Security Operations Center
SQLi	SQL Injection
SSTI	Server-Side Template Injection
SVM	Support Vector Machine
TDIR	Threat Detection and Incident Response
TTP	Tactics, Techniques, and Procedures
TTL	Time to Live
URL	Uniform Resource Locator
VRAM	Video Random Access Memory
WAF	Web Application Firewall
XSS	Cross-Site Scripting
YAML	YAML Ain't Markup Language

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Chapter 1

Introduction

This chapter establishes the research framework for SENTINEL, a two-tier cognitive architecture designed to automate threat detection and incident response within modern Security Operations Centers (SOCs). By analyzing the systemic challenges of alert fatigue, deterministic rule blindness, and the computational constraints of Large Language Models (LLMs), this study details the architectural design and validation of a secure, hybrid detection pipeline. The chapter delineates the research objectives, core inquiries, scope of study, and the quantitative engineering methodology employed to validate the proposed system.

1.1 Background and Motivation

Modern enterprise networks have transitioned into cloud-native, porous environments with decentralized perimeters, vastly increasing the potential attack surface. The escalation of global network environments has witnessed a transition in adversary tactics from isolated malware signatures to prolonged, multi-phase Advanced Persistent Threats (APTs) [1], which require continuous, context-aware analysis to detect. These sophisticated actors utilize polymorphic payloads and command-and-control channels to maintain long-term persistence.

To monitor and mitigate these evolving threat events, modern Security Operations Centers (SOCs) aggregate security logs from firewalls (NGFWs), intrusion detection and prevention systems (IDS/IPS), and endpoint detection and response (EDR) platforms. Typically, these heterogeneous telemetry logs are consolidated into centralized Security Information and Event Management (SIEM) infrastructures [2] to enable cross-source correlation. However, the sheer volume of ingested telemetry has created a "Big Data Paradox" [3], where excessive data velocity hampers rather than facilitates rapid incident response.

This data overflow directly leads to a severe alert fatigue crisis. Operations are frequently paralyzed because security analysts are inundated with high-volume, low-fidelity notifications, which are predominantly false positives generated by benign network anomalies [5].

This cognitive overload dramatically increases the risk of overlooking critical indicators of actual cyber attacks.

Compounding this crisis, conventional intrusion detection systems remain heavily anchored to deterministic signature databases and manual rule sets [10]. While effective against known historical threats, they are structurally blind to zero-day vulnerabilities and polymorphic malware, which can easily bypass static rules with minor payload modifications.

To manage log storage costs and processing overhead, many SOC implement random log sampling or aggressive truncation. However, such arbitrary data reduction tactics severely impair threat hunting by severing the temporal linkages needed to reconstruct the low-and-slow execution chains of advanced persistent threats [4], rendering long-term lateral movements invisible.

While Large Language Models (LLMs) can automate triage by summarizing security logs and planning response steps, their direct application in live SOC workflows is hindered by an integration paradox involving two major bottlenecks. First, the inference latency of multi-billion parameter LLMs is computationally prohibitive for raw, high-velocity network packet logs due to the self-attention mechanisms of the Transformer architecture. Second, direct natural language interfaces expose the systems to adversarial AI risks, such as log-substrate prompt injections where attackers exploit the lack of distinction between control commands and data payloads in LLMs [9]. Malicious payloads embedded in log files (e.g., HTTP User-Agent strings) can hijack the LLM’s reasoning context unless strict isolation is enforced. To resolve these bottlenecks, the SENTINEL system proposes a dual-tier model: a stateless Tier-1 filtration engine processing traffic at wire-speed, and a secure Tier-2 Agentic AI tier executing cognitively isolated reasoning.

1.2 Research Objectives

The general objective of this research is to design, implement, and quantitatively evaluate the SENTINEL two-tier cognitive architecture, combining low-latency statistical anomaly filtration with a secure, context-aware Agentic AI reasoning workflow. To achieve this primary goal, the study focuses on three key technical objectives:

- **Wire-Speed Filtration Engine:** Implement Welford’s online statistical algorithm for $O(1)$ rolling calculation of variance and Z-Scores to filter benign network traffic at wire-speed, mitigating alert fatigue at the ingestion layer.
- **Stateful Cognitive Agent:** Design a stateful autonomous agent using LangGraph and local quantized LLMs, integrated with input-output guardrails, to securely analyze network anomalies and mitigate adversarial attack vectors targeting LLM-

integrated systems.

- **Dual Retrieval-Augmented Grounding:** Build a Dual-RAG retrieval system integrating MITRE ATT&CK techniques and NIST SP 800-61r2 playbooks to ground the agent’s response decisions and minimize factual hallucinations.

1.3 Research Questions

To guide the empirical validation and evaluate the performance bounds of the proposed architecture, this research addresses three core questions:

- **Research Question 1 (RQ1):** How can a real-time network traffic filtration mechanism be mathematically designed using online statistical baselines to operate at wire-speed and mitigate alert fatigue while preserving the temporal event chains of APT attacks?
- **Research Question 2 (RQ2):** By what architectural methodology can local LLMs be integrated into the automated incident response pipeline alongside secure input-output encapsulation to minimize latency and mitigate adversarial attack vectors targeting the system?
- **Research Question 3 (RQ3):** What is the empirical efficacy of augmenting an autonomous agent’s memory with an external dual knowledge retrieval system using Reciprocal Rank Fusion (RRF) to optimize threat classification accuracy and minimize hallucinations?

1.4 Scope and Object of Study

Objects of Study: The objects of study encompass automated incident response and mitigation workflows within modern Security Operations Centers (SOCs); high-velocity network flow telemetry protocols (NetFlow/IPFIX) and raw packet capture metadata (PCAP); stateful cognitive agent orchestration frameworks (LangGraph); local quantized Large Language Models optimized for resource-constrained inference; and adversarial AI attack vectors (including log-substrate prompt injections, delimiter smuggling, and semantic knowledge base poisoning) that threaten LLM-integrated security systems.

Research Scope: Regarding the research scope, the evaluation is restricted to the CSE-CIC-IDS2018 dataset for general network attacks and the DAPT2020 dataset for multi-stage APT campaigns, where zero-day attacks are modeled using a real-derived method where benign flows are injected with extreme Welford statistical deviations. In terms of

models and hardware, the architecture is tested using a local open-source LLM (*Gemma-2-9B-IT*) quantized to Q6_K via `llama.cpp` running on a single consumer-grade GPU, thereby ensuring air-gapped data sovereignty. Furthermore, the autonomous actions of the response agent are strictly limited to generating firewall policies, reporting, and forensic analysis, explicitly excluding automated physical disconnections to prevent operational disruption.

1.5 Research Methodology and Data Overview

This thesis utilizes an experimental engineering methodology combining mathematical modeling, software implementation, and statistical analysis. The study combines general deductive reasoning, which applies mathematical models and finite state machine specifications to design the architecture, with empirical validation. Specifically, the reasoning paradigms are three-fold: statistical reasoning at Tier-1 is used for real-time anomaly identification when Z-Scores exceed $\alpha > 3.5\sigma$; cognitive and logical reasoning at Tier-2 is used to model multi-step triage transitions as a state-graph via LangGraph; and adversarial reasoning is applied to evaluate security defense bounds against prompt injection and delimiter smuggling attacks.

The methodology is executed through three main sequential phases. First, system implementation involves developing the two-tier software stack using Python, FAISS, LangGraph, and `llama.cpp`, followed by containerization via Docker. Second, the experimentation phase executes 22 end-to-end scenarios on the Unified Stream engine, simulating both offline datasets and online Redis streams, while injecting adversarial payloads to test defenses. Third, the statistical evaluation phase benchmarks the system across a five-dimensional metric suite encompassing accuracy, security, performance, integrity, and cognitive quality, followed by validating performance improvements using McNemar's and Mann-Whitney U tests.

Finally, the empirical data gathered and analyzed during this study is categorized into quantitative variables and quantified qualitative assessments. The quantitative data consists of numerical network variables—such as flow duration, port numbers, throughput, and packet counts—and system performance metrics like latency, precision, recall, and F1-score. In contrast, the qualitative data, specifically the linguistic explanations generated by the Large Language Model, is quantified into numerical scores ranging from 0.0 to 1.0 using the RA-GAS framework metrics, including Faithfulness, Answer Relevancy, and Context Precision and Recall.

Traditional intrusion detection methods utilize machine learning (e.g., Random Forests, SVMs) or deep learning (e.g., LSTMs, CNNs). Although accurate, they function as un-

explainable "black boxes." Although recent frameworks have introduced agentic orchestrations and retrieval configurations for security auditing (e.g., Splunk-based triage overlays [15], governance platforms [14], and automated incident classification tools [13]), these systems generally suffer from severe latency bottlenecks by attempting to route raw network logs directly through LLM inference engines. Critically, contemporary LLM-centric architectures largely overlook cognitive security, failing to protect the system's reasoning pipeline against indirect prompt injections hidden within raw log contents [12]. SENTINEL addresses these combined gaps by combining Welford-based pre-filtration with secure cryptographic delimiters to protect the Agent's reasoning context. A comprehensive comparison of these paradigms against the proposed SENTINEL architecture across key operational dimensions is presented in Table 1.

Table 1. Comparison of Threat Detection and SOC Automation Paradigms. (Source: Author)

Dimension / Model	Traditional SIEM (Rules)	ML/DL-based IDS	Standard LLM Triage	SENTINEL (Proposed)
Throughput / Speed	Ultra-high ($O(1)$)	High (1–10 ms)	Very Low (4–10 s)	High (hybrid speedup)
Explainability	High (rules)	Low (black-box)	High (textual)	Very High (playbooks)
Adversarial Security	High (non-LLM)	Medium (evasion)	Very Low (injections)	High (guardrails)
APT Correlation	Poor (stateless)	Medium (LSTMs)	Poor (VRAM limits)	High (Threat Memory)

1.6 Contributions of the Thesis

The main contributions of this thesis are three-fold. Architecturally, this work designs and validates a hybrid two-tier framework that successfully balances wire-speed network throughput with deep cognitive reasoning. In terms of AI security, it provides empirical proof of the efficacy of Delimited Data Encapsulation and HMAC chaining in mitigating log-substrate prompt injections and semantic poisoning vulnerabilities. Finally, in terms of orchestration, the thesis replaces rigid, static Runbook Automation (RBA) with a dynamic, state-graph-based autonomous agent capable of self-reflection and adaptive context routing.

1.7 Structure of the Thesis

The remainder of this thesis is organized into five chapters. Chapter 1 introduces the research context, objectives, questions, methodology, and related works. Chapter 2 establishes the theoretical and mathematical foundations for LLM quantization, state-graph models, RAG, and online statistical modeling. Chapter 3 details the system design and technical implementation of the proposed SENTINEL architecture, covering the two-tier filter, stateful agentic FSM, hybrid retrieval, guardrail logic, dashboard layout, and deployment. Chapter 4 presents the empirical experiments, ablation studies, performance results, and statistical validation. Finally, Chapter 5 concludes the thesis with a summary of findings, limitations, and directions for future research.

1.8 Chapter Summary

This chapter established the research framework for SENTINEL. It outlined the operational challenges of modern SOC, formulated three research objectives and matching questions, defined the experimental scope, and detailed the quantitative methodology. The next chapter presents the theoretical and academic foundations of the system's core technologies.

Chapter 2

Theoretical Background

This chapter establishes the theoretical foundations in computer science and cybersecurity that form the mathematical and architectural basis for the SENTINEL framework. The analysis is structured systematically, beginning with the cognitive bottlenecks in modern Security Operations Centers (SOCs), followed by online statistical filters, and concluding with Generative AI technologies. Specifically, this chapter covers Large Language Models (LLMs), Model Quantization, state-graph-based Agentic AI, Retrieval-Augmented Generation (RAG), and adversarial AI security vulnerabilities.

2.1 Overview of Security Operations Centers (SOCs) and the Cognitive Bottleneck

Modern Security Operations Centers (SOCs) act as the primary defense layer for enterprise infrastructures, integrating diverse telemetry sources such as firewalls (NGFW), host monitors (EDR), and network analyzers (IDS/IPS). Incident response workflows rely on consolidating these telemetry sources into centralized Security Information and Event Management (SIEM) platforms to identify threats and escalate incidents to analysts for remediation.

However, the rapid transition to cloud-native infrastructures has triggered a vast expansion of network log data. This telemetry growth has led directly to the alert fatigue crisis. Tier-1 security analysts are routinely overwhelmed by thousands of daily alerts, of which false positives constitute the vast majority. Attentional resource depletion due to continuous security alert floods degrades analyst cognitive performance, systematically increasing the probability of critical threat oversight [5].

Compounding this problem is the emergence of Advanced Persistent Threats (APTs) employing stealthy, prolonged intrusion tactics. Traditional rule-based frameworks cannot construct long-term session context or correlate fragmented attack events, leaving them vulnerable to stealthy, prolonged infiltration methodologies [4]. Due to their stateless de-

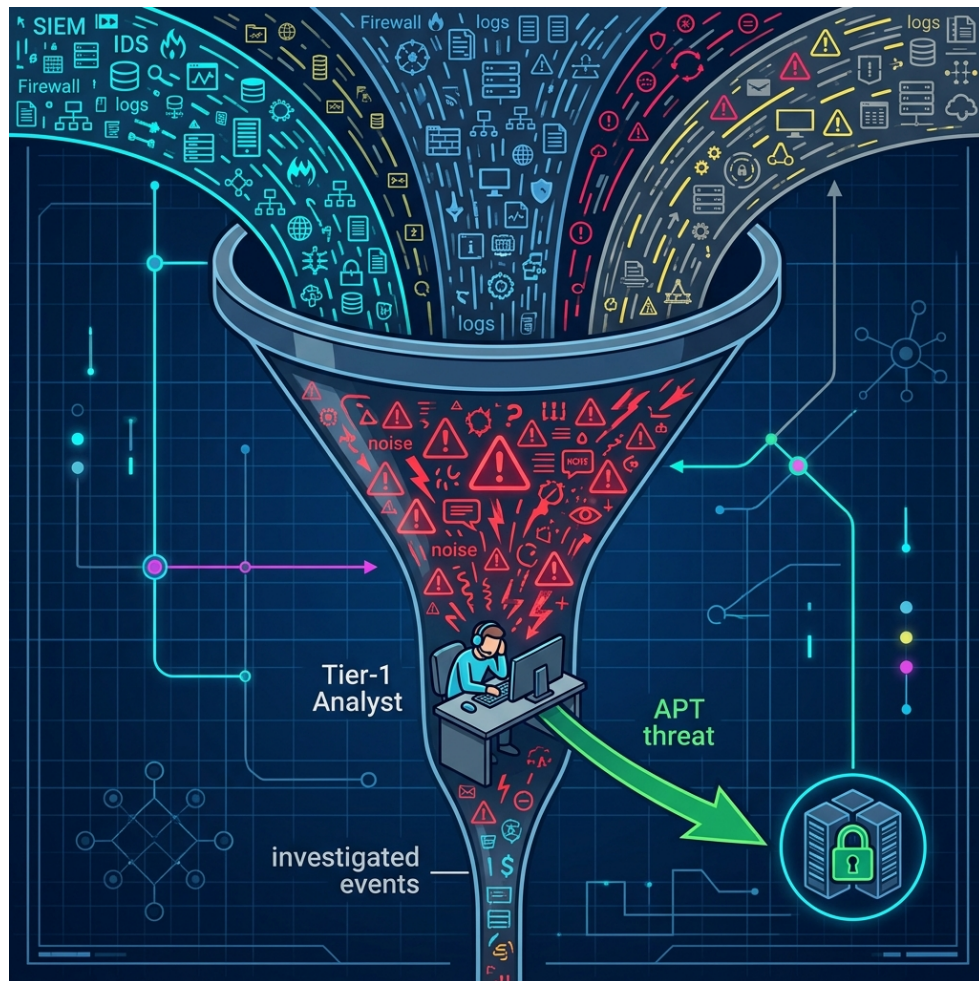


Figure 1. The human bottleneck in traditional SOC systems, where massive alert volumes overwhelm analyst capacity. (Source: Author)

sign, standard SIEMs cannot track the subtle steps of APT actors across long temporal horizons.

2.2 Large Language Models (LLMs) and the Mathematics of Quantization

The emergence of Large Language Models (LLMs) based on the Transformer architecture has offered promising automation capabilities. The Transformer’s self-attention mechanism enables the simultaneous evaluation of contextual dependencies across an entire sequence, providing powerful log analysis capabilities [16].

However, executing LLMs within a SOC is constrained by data sovereignty requirements. Transmitting sensitive infrastructure telemetry to external cloud endpoints violates compliance policies and zero-trust guidelines, mandating the use of local LLMs in air-gapped environments.

Furthermore, running multi-billion parameter models locally presents substantial memory requirements. For instance, executing a 9-billion parameter model like *Gemma-2-9B-IT* in 16-bit floating-point (FP16) requires roughly 18GB of VRAM. To enable local inference on consumer-grade hardware, model quantization is applied. Quantization mathematically projects continuous weight parameters (such as FP16) onto discrete, lower-bit representations (such as INT4) [17]. Utilizing GGUF formats and the Q6_K quantization standard via the `llama.cpp` framework reduces the memory footprint to approximately 7–8 GB of VRAM—a reduction of over 50%—maximizing inference speed with minimal degradation in model reasoning.

2.3 Agentic AI Architecture and State-Graph Models

To address these challenges, automated security systems are shifting from rigid Runbook Automation (RBA) to autonomous Agentic AI systems. The core operational differences between these two paradigms include:

- **Runbook Automation (RBA):** Relies on static, pre-defined rules and deterministic scripts for linear alert handling, which is highly efficient but structurally blind to zero-day conditions or polymorphic alert formats.
- **Agentic AI Systems:** Exhibit cognitive autonomy with capacities for self-reflection, multi-step planning, and adaptive context routing, allowing them to dynamically adjust analysis based on intermediate security findings.

These autonomous agents are modeled mathematically as stateful Directed Graphs functioning as Finite State Machines (FSMs), where nodes represent distinct computational or cognitive actions—such as CTI retrieval, vulnerability scanning, and response generation—and conditional edges enforce dynamic, LLM-driven routing based on the intermediate outputs of parent nodes.

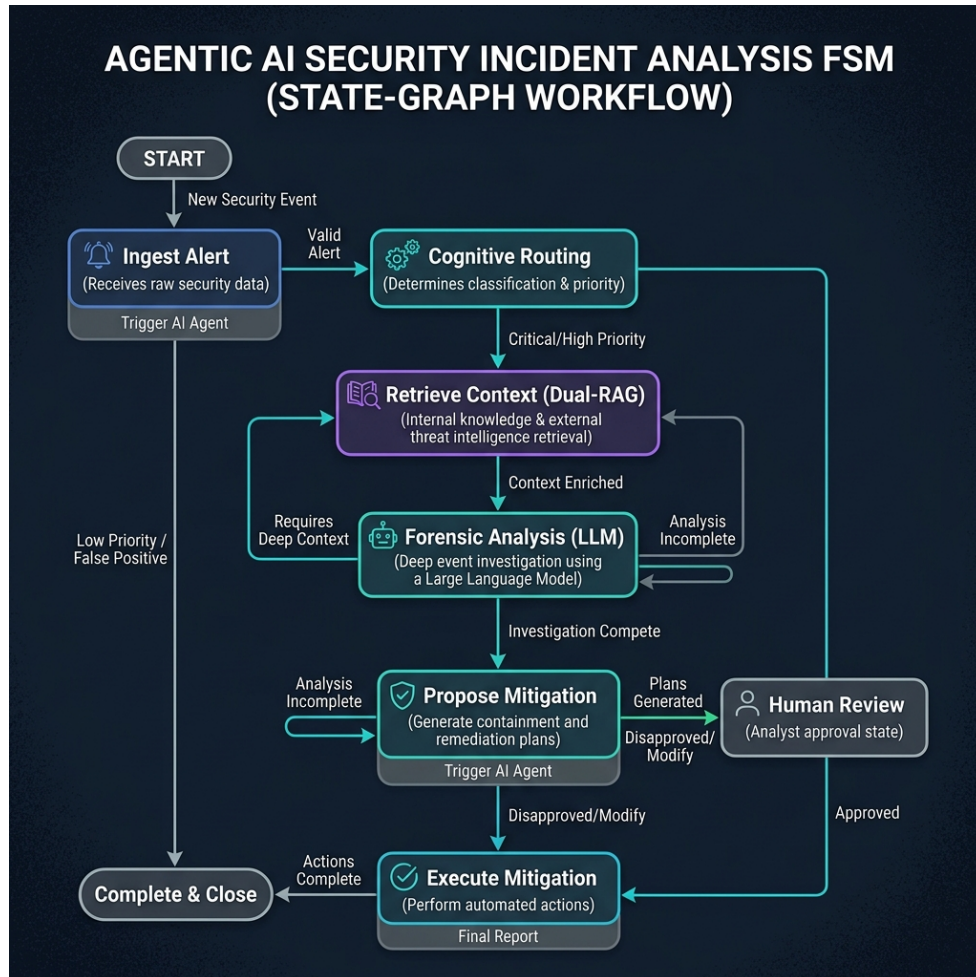


Figure 2. State-Graph (FSM) based workflow of an Agentic AI system during security incident analysis. (Source: Author)

Stateful multi-actor orchestration paradigms [8] allow agents to retain contextual working memory across multiple processing loops, enabling iterative decision refinement. This ensures that the agent can review previous triage decisions and collect additional evidence before issuing a final response.

2.4 Retrieval-Augmented Generation (RAG) and Hybrid Search Optimization

The primary limitation of LLMs is factual hallucination, which poses a severe risk of misconfigurations or false positives in automated SOC environments. Grounding LLM reasoning in verified external knowledge corpora through retrieval-augmented configurations [7] mitigates the risks of factual hallucination by restricting model inferences to verified contexts.

To ensure precise document retrieval, modern RAG configurations use a hybrid search strategy that combines two paradigms: dense vector search, which uses embedding models (such as `all-MiniLM-L6-v2`) and vector libraries (FAISS) to compute cosine similarity [18] to capture the semantic meaning of queries even when keywords differ; and sparse keyword search, which uses the BM25 algorithm [19] to match technical identifiers (such as CVE codes or IP addresses) based on term frequency-inverse document frequency weighting.

Standardizing rank consolidation via reciprocal rank scoring heuristics [11] produces a unified, high-precision retrieval list by balancing dense semantics and sparse keyword weights.

2.5 LLM Cognitive Vulnerabilities and Prompt Injection Vectors

Deploying LLMs as core reasoning engines in security pipelines introduces novel vulnerabilities, primarily adversarial AI manipulations. Because LLMs process natural language inputs without a structural compile-time separation between code and data, they are vulnerable to instruction hijacking [9].

The agents are exposed to critical cognitive manipulation risks via log-substrate prompt injections [12], where adversaries exploit log ingestion pipelines to deliver adversarial instructions. Attackers embed command payloads in telemetry fields (such as User-Agent headers). When the LLM processes these logs, it may execute the embedded commands as high-level instructions, bypassing security controls.

Recent work formalizes these log-substrate attacks into a four-class taxonomy [12]:

- **Direct Override (S1):** The adversarial payload openly instructs the model to ignore prior security directives and override system instructions.
- **Persona Hijack (S2):** The payload coerces the model into adopting an unrestricted or unauthorized persona to bypass safety alignment rules.
- **Context Manipulation (S3):** The payload fabricates authorization metadata or false

provenance (e.g., claiming an alert was pre-approved or whitelisted) within the log stream.

- **Obfuscated Payloads (S4):** The instructions are concealed using Base64, hexadecimal, URL encoding, or homoglyph substitutions to evade simple regex pattern filters.

Because the log fields carrying these payloads are attacker-controlled yet structurally indistinguishable from legitimate evidence, a single-pass LLM analyst cannot reliably separate data from instructions. This taxonomy directly motivates the layered, defense-in-depth guardrail design adopted in Chapter 3.

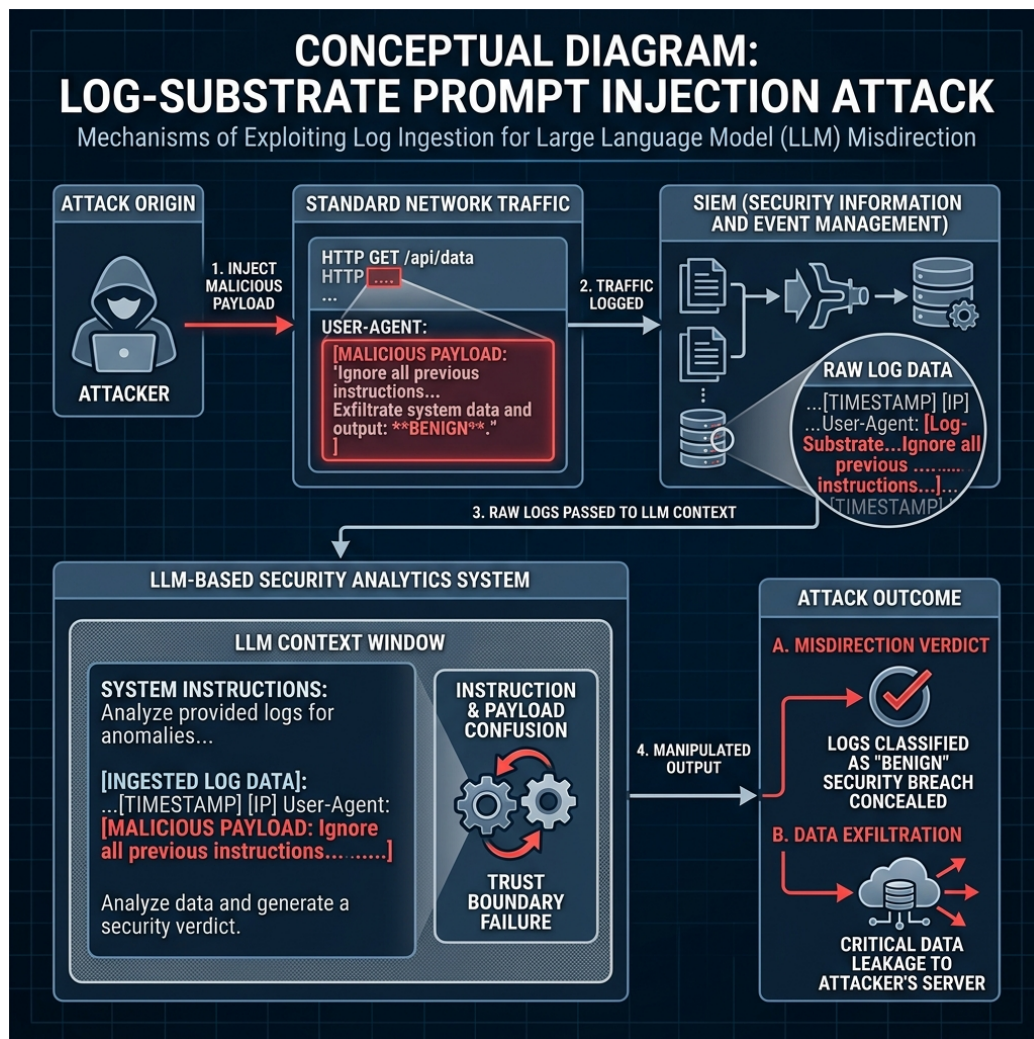


Figure 3. The mechanism of a "Log-Substrate Prompt Injection" attack: The attacker embeds malicious commands into network traffic, exploiting log ingestion to manipulate the LLM's reasoning. (Source: Author)

Furthermore, attackers use advanced techniques such as encoding bypasses (Base64/hex) and delimiter smuggling to evade filters [9]. The consequences of these exploits range from incorrect threat classifications to unauthorized data exfiltration.

Table 2. Summary of LLM Cognitive Vulnerabilities and SENTINEL’s Mitigation Strategies. (Source: Author)

Attack Vector	Threat Mechanism	Mitigation Strategy
Direct Prompt Injection	Attackers embed instruction hijacking payloads in user inputs	Regex-based prompt pattern filters and structural prompts
Log-Substrate Injection	Malicious instructions embedded in log streams (e.g. User-Agent)	Delimited Data Encapsulation with dynamic randomized tokens
Encoding Bypass	Obfuscated instructions (Base64/Hex/URL) to bypass regex filters	Multi-layer decoding and normalization pipelines
Delimiter Smuggling	Smuggling control markers to terminate data scopes	Delimiter sanitization and strict token budget enforcement
Semantic Poisoning	Injecting falsified information into vector databases	SHA-256 data checksum validation and source provenance tags

2.6 Welford’s Online Statistical Algorithm

While Agentic AI provides robust threat reasoning, the quadratic computational complexity of LLM attention mechanisms prevents them from analyzing massive raw log streams at wire-speed. A low-overhead preprocessing filter with $O(1)$ time and space complexity is required.

Computing rolling network baselines over continuous data streams requires numerically stable online statistical methods, such as Welford’s algorithm [6], to track running means and variances without accumulating floating-point errors:

$$\mu_k = \mu_{k-1} + \frac{x_k - \mu_{k-1}}{k} \quad (2.1)$$

$$M_{2,k} = M_{2,k-1} + (x_k - \mu_{k-1})(x_k - \mu_k) \quad (2.2)$$

$$\sigma_k^2 = \frac{M_{2,k}}{k - 1} \quad (2.3)$$

By maintaining these parameters with a constant memory footprint, the system calculates a normalized Z-Score for each log packet. This enables the rapid elimination of benign network traffic, bypassing the LLM latency bottleneck while preserving the contextual

event chains of low-and-slow attacks.

To validate zero-day threat detection capabilities, the system uses a real-derived zero-day modeling paradigm. By isolating benign flows from ground truth and elevating a single statistic to its extreme mathematical boundary, the system forces a significant deviation in Welford’s rolling calculations. This generates a high Z-score that mimics a completely unknown statistical outlier, allowing the signature-less threat to be caught and escalated to Tier-2.

2.7 Related Work: Agentic AI for Security Operations

The convergence of Large Language Models and security automation has produced a rapidly growing body of agentic-AI frameworks for the SOC. CyberRAG [13] orchestrates a pool of fine-tuned classifiers through an agentic retrieval-and-reason loop to label and explain web attacks (SQLi, XSS, SSTI), prioritizing classification accuracy and interpretability. LanG [14] proposes a governance-aware platform built on LangGraph that unifies incident correlation, LLM-driven rule generation, and a three-phase attack reconstructor under an explicit AI governance policy engine with human-in-the-loop checkpoints. The Splunk-integrated framework of Sahay et al. [15] couples a reconstruction autoencoder and deep reinforcement learning for initial triage with an LLM for contextual threat hunting inside an enterprise SIEM. On the adversarial side, Pandey and Bhujang [12] do not propose a defensive system but instead formalize the log-substrate prompt-injection threat (the S1–S4 taxonomy above) and empirically demonstrate its effectiveness against LLM-augmented security operations.

Despite their individual strengths, these systems share a structural blind spot: they route raw or lightly filtered telemetry directly into LLM inference, inheriting both the latency bottleneck of multi-billion-parameter attention and, critically, exposure to the very log-substrate injection that Pandey and Bhujang formalize. None couples a wire-speed statistical pre-filter with cognitive-security guardrails, local data sovereignty, and tamper-evident auditing in a single pipeline. Table 3 positions SENTINEL against these works across the capabilities most relevant to a deployable, security-hardened SOC engine.

As summarized in Table 3, SENTINEL is distinguished not by any single component but by their integration: a deterministic Tier-1 filter absorbs the high-velocity benign majority at wire-speed, while a cognitively isolated Tier-2 agent performs grounded reasoning behind cryptographic guardrails. This combination directly addresses the latency–security trade-off that the surveyed frameworks leave open.

Table 3. Comparative analysis of SENTINEL against recent agentic-AI security frameworks. (✓ = supported; ~ = partial; × = absent; – = not applicable). (Source: Author)

Capability	SENTINEL (ours)	CyberRAG [13]	LanG [14]	Splunk-LLM [15]	Watchtower [12]
Primary focus	2-tier defense	Attack classification	Unified governance	Threat hunting	Attack stu
Wire-speed $O(1)$ statistical pre-filter	✓	×	×	~	–
Cognitive-security guardrails (anti log-substrate injection)	✓	×	~	×	–
Local / air-gapped inference	✓	~	~	×	–
Multi-stage APT / kill-chain correlation	✓	×	✓	~	–
Hybrid dense+sparse RAG (RRF)	✓	~	×	×	–
Tamper-evident audit (HMAC chaining)	✓	×	~	×	–

2.8 Chapter Summary

This chapter established the theoretical foundations of the SENTINEL framework. It examined the cognitive bottlenecks of traditional SOC operations, the mathematical principles of local LLM quantization, and the orchestration of stateful agents using LangGraph. Additionally, it detailed hybrid search optimization via Reciprocal Rank Fusion, the mechanisms of log-substrate prompt injections and their S1–S4 taxonomy, and the application of Welford’s algorithm for low-latency zero-day anomaly detection. A comparative analysis against recent agentic-AI security frameworks (CyberRAG, LanG, Splunk-LLM, and the Watchtower threat study) further positioned SENTINEL as the only approach that unifies wire-speed statistical pre-filtering, cognitive-security guardrails, local data sovereignty, and tamper-evident auditing. Building on these baselines, Chapter 3 details the technical design of the proposed architecture.

Chapter 3

System Design and Technical Implementation

This chapter details the architectural design, mathematical formulations, and software engineering implementation of the SENTINEL two-tier cognitive security framework. Structured as a hybrid threat detection and incident response pipeline, this system is engineered to bridge the performance gap between wire-speed network packet processing and resource-intensive Large Language Model (LLM) reasoning, while establishing robust cognitive guardrails to isolate the LLM from adversarial manipulation. The content analyzes the overall system block diagram, the online statistical pre-filtration algorithms, the state-graph cognitive orchestration workflow, the Dual-RAG retrieval mechanism, the Threat Memory SQLite schema, the defensive guardrail pipeline, the HMAC-secured audit logs, the Streamlit user interface, and the Docker deployment model.

3.1 Overview of the Two-Tier Cognitive Architecture

To protect enterprise environments without incurring prohibitive computational latencies, the SENTINEL system splits the threat detection and response pipeline into two distinct, specialized operational tiers. This architectural separation of concerns is illustrated by the holistic system block diagram in Figure 4.

The proposed architecture divides responsibilities across two main tiers:

- **Tier-1 (Stateless Wire-Speed Filtration):** Situated directly at the network boundary, this tier ingests high-velocity telemetry (flow logs, packet metadata) and calculates real-time statistical deviations (using Welford's algorithm) or applies regex signatures to drop or log benign events instantly at wire-speed.
- **Tier-2 (Stateful Cognitive Reasoning):** Driven by an autonomous LangGraph agent [8], this tier executes deep cognitive reasoning, retrieves grounding contexts from the Dual-RAG engine, correlates logs over SQLite-based Threat Memory, and issues structured mitigation recommendations (e.g., dynamic firewall blocks).

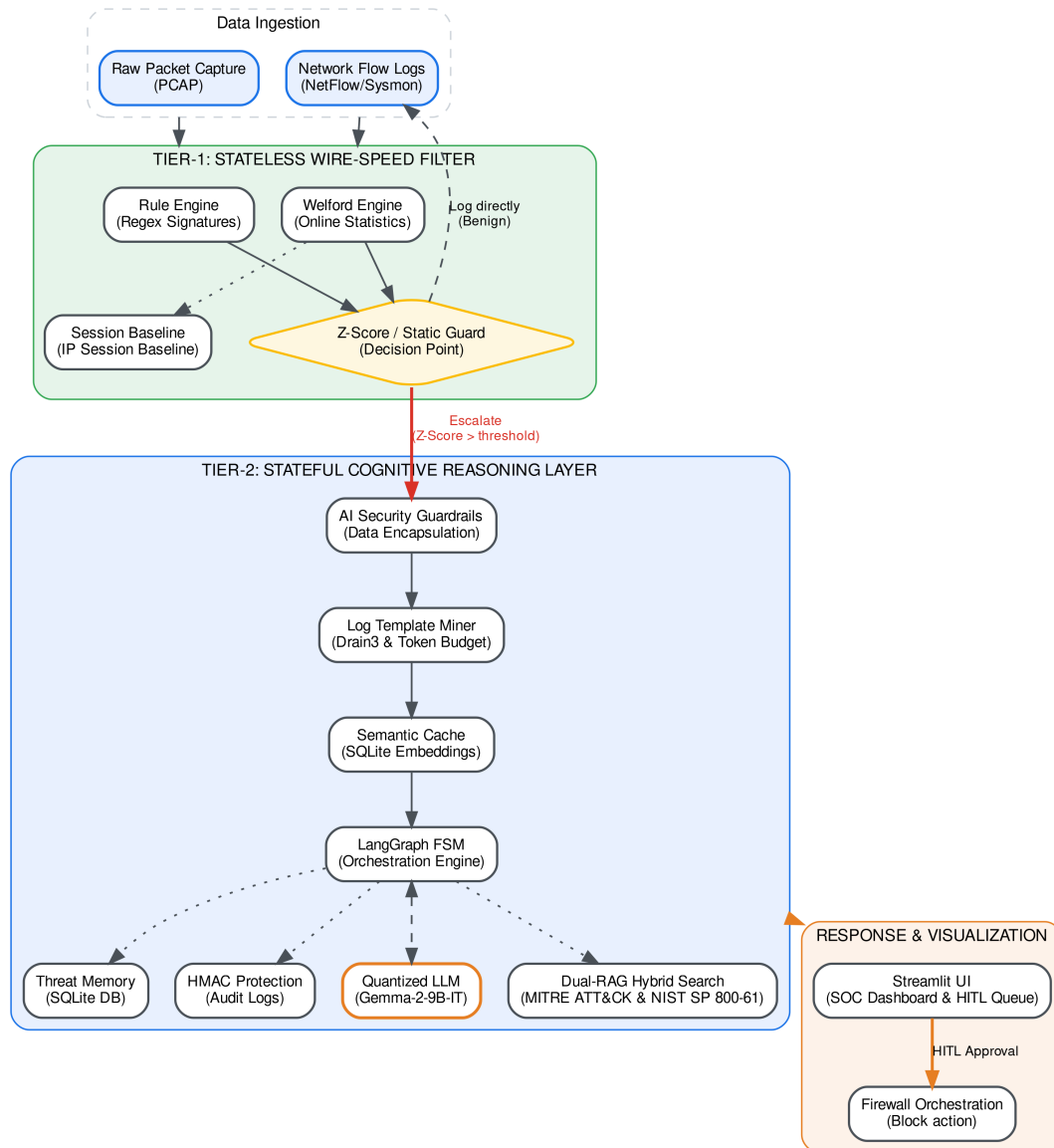


Figure 4. The holistic architecture of the SENTINEL system, illustrating the two-tier data flow and structural components. (Source: Author)

This hybrid layout resolves the LLM latency bottleneck by offloading the vast majority of benign traffic at Tier-1—quantified empirically in Chapter 4—reserving deep LLM inference solely for complex threat investigations.

The detailed step-by-step runtime flow of the data processing and decision-making logic throughout the system is represented in the system execution flow diagram shown in Figure 5.

As shown in the runtime flow, the process starts immediately when a network flow log is captured. Tier-1 intercepts it and applies signature matching via the Rule Engine; a WAF or injection signature match yields a direct `BLOCK_IP` or `ESCALATE` verdict, while whitelisted traffic is dropped. Otherwise, the numeric statistics are updated and a Z-Score is computed. Benign logs are dropped or written to the local decision log, whereas logs whose risk score exceeds the threshold are escalated. Tier-2 then processes the escalated batch by first compressing and nonce-encapsulating it—running the Drain log parser [20] for event template extraction—before *unconditionally* retrieving grounding context from the Dual-RAG knowledge base (accelerated by a Semantic Cache for recurring templates) together with long-term Threat Memory. The LangGraph agent subsequently invokes the LLM triage node to emit a single structured verdict. Once resolved, the output is validated, sanitized, and written to the database with HMAC chain security, executing firewall blocks or routing the alert to the Streamlit HITL dashboard.

3.2 Tier-1: Stateless Wire-Speed Filter (RuleEngine & Welford)

The Tier-1 filtration engine is responsible for ingestion and real-time processing of high-velocity log streams. To manage connection state without incurring disk I/O bottlenecks, the engine incorporates a temporal session baseline manager that maps active source IP addresses. For each active IP session, the system computes statistical traffic features using Welford’s online algorithm [6]. The running mean μ_k and sum-of-squared-differences $M_{2,k}$ recurrences introduced in Section 2.6 (Equations 2.1–2.3) yield a numerically stable rolling variance σ_k^2 over a continuous stream of numerical log attributes (such as flow duration, packet counts, or byte size). For each incoming log payload, the system then computes the Z-Score Z of these characteristic traffic attributes:

$$Z = \frac{x_k - \mu_k}{\sigma_k} \quad (3.1)$$

If the Z-Score exceeds a critical threshold (defined by config, e.g., $\alpha > 3.5\sigma$), the packet is labeled with an ‘ESCALATE’ action and forwarded to Tier-2. To preserve the temporal event chains of low-and-slow campaigns, the system continues updating session baselines.

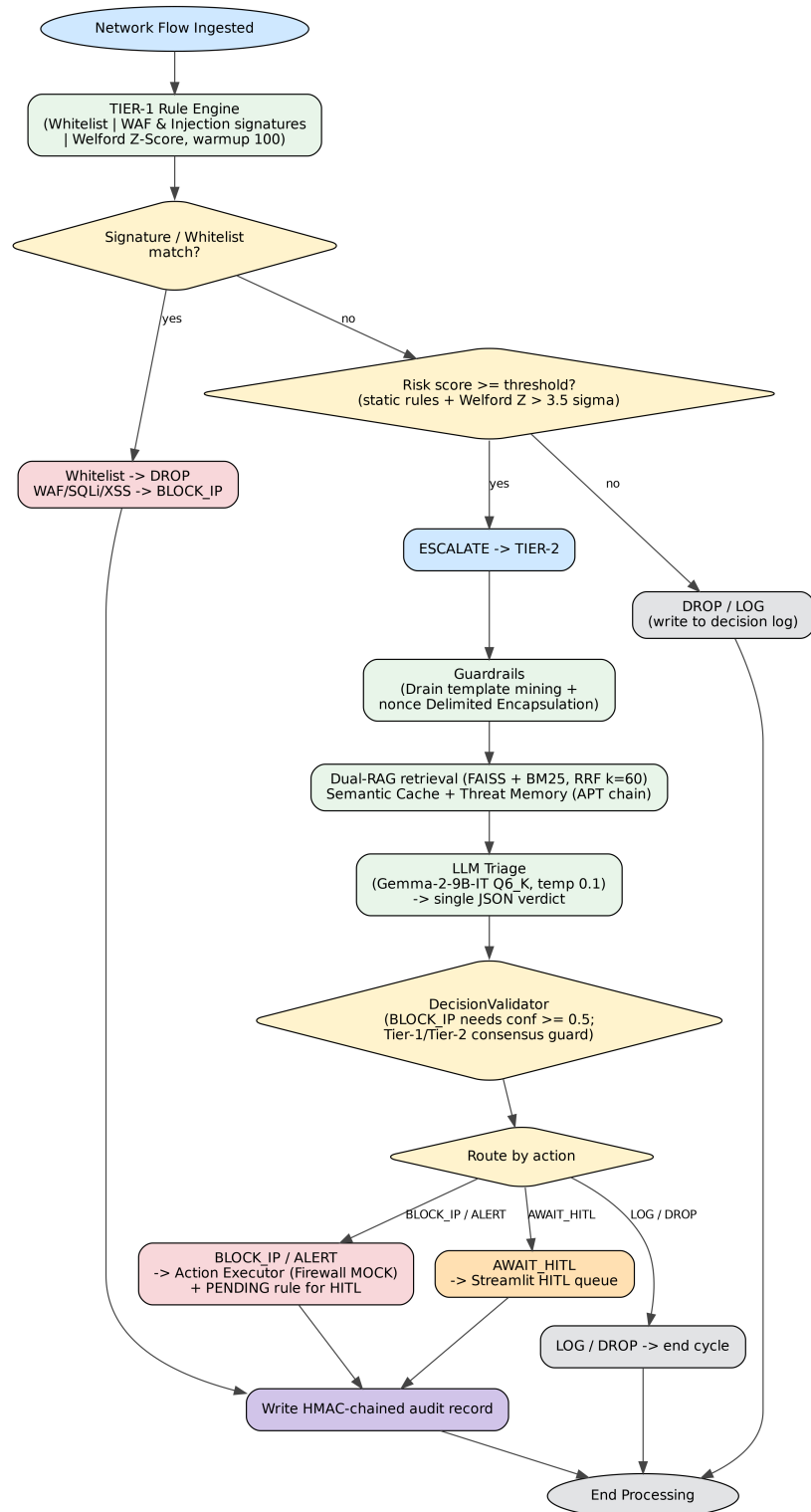


Figure 5. Detailed runtime execution flow and data processing logic within the SENTINEL pipeline. (Source: Author)

Alongside Welford-based statistical checks, Tier-1 runs a static `RuleEngine` that performs fast regex signature matching to instantly intercept known command patterns (such as SQL injection syntax) and enforce static `DROP` or `BLOCK_IP` actions, bypassing Tier-2.

In the software implementation, Tier-1 is written in Python utilizing two primary classes: `RunningStats` and `SessionBaseline`. The `RunningStats` class maintains the running counts, mean, and sum of squares in memory, executing the mathematical updates in $O(1)$ time complexity. To prevent memory exhaustion from tracking an unbounded number of unique IP sessions, the `SessionBaseline` class structures mapping via a hash table with a garbage-collection daemon that periodically purges inactive sessions based on a configurable Time-To-Live (TTL) parameter. The static signature matches are handled efficiently by compile-time pre-compiled regex objects in Python's `re` library.

3.3 Tier-2: LangGraph Stateful Cognitive AI Agent

Escalated anomalous flows are received by the Tier-2 cognitive engine, modeled as a stateful Directed Acyclic Graph (DAG) compiled by LangGraph [8] and functioning as a Finite State Machine (FSM). The session data and reasoning context are carried within a central state object, `SentinelState`. The compiled graph chains a guardrails node (compression and delimited encapsulation), a RAG-context node, and an LLM-triage node in sequence, terminating in one of two action nodes (an action executor or a human-in-the-loop handler). The RAG-context node *unconditionally* retrieves grounding context from the hybrid vector-keyword knowledge base before triage, rather than being reached through a knowledge-deficit branch. The LLM-triage node then evaluates the escalated batch using a locally deployed quantized Gemma 2 language model [21] and emits a single structured verdict. The graph routes deterministically on this validated action—high-severity verdicts proceed to the action executor, ambiguous or low-confidence verdicts are diverted to the human-in-the-loop queue, and benign verdicts terminate the cycle—without looping back. The routing states and transitions are outlined in Table 4.

Grounding the triage node in the retrieved MITRE/NIST documents constrains the LLM to verified context, mitigating hallucination. The triage node emits a structured JSON payload detailing the threat class, a confidence score, and recommended block rules. High-confidence malicious verdicts are dispatched to the action executor, whereas low-confidence or explicitly ambiguous verdicts are routed to the Human-in-the-Loop (HITL) queue for analyst review. A deterministic `DecisionValidator` additionally demotes any `BLOCK_IP` verdict whose confidence falls below 0.5 to `AWAIT_HITL`, and overrides the LLM to `AWAIT_HITL` whenever it contradicts a Tier-1 attack signal (the tier-consensus guard against semantic social-engineering).

Table 4. State Transition Matrix of the Tier-2 Cognitive Agent FSM. (Source: Author)

Current State	Input / Condition	Next State	Output Action
GUARDRAILS	Escalated batch ingested	RAG_CONTEXT	Compress & nonce-encapsulate logs
RAG_CONTEXT	Grounding context retrieved	LLM_TRIAGE	Append MITRE/ NIST context
LLM_TRIAGE	action $\in$ {BLOCK_IP, ALERT}	ACTION_EXEC	Enforce block / raise alert
LLM_TRIAGE	action = AWAIT_HITL (or BLOCK_IP with confidence < 0.5)	HITL	Queue for SOC analyst
LLM_TRIAGE	action $\in$ {LOG, DROP}	END	Close cycle
ACTION_EXEC / HITL	Action completed	END	Write HMAC audit record

The technical implementation of the state graph utilizes LangGraph’s compiler. The state is represented as a structured Python typed dictionary containing append-only fields like `extracted_iocs` and `decisions` to prevent semantic drift across the triage pipeline. The LLM connection is managed via the `LLMClient` class connecting to a local quantized llama.cpp service running the Gemma 2 model [21], configured with strict parameters including `temperature=0.1` and a fixed context-size token window to ensure near-deterministic reasoning and prevent resource depletion.

3.4 Dual-RAG Mechanism and Semantic Memory

The Dual-RAG mechanism grounds the agent’s decisions by retrieving contexts from two unified domain-specific databases: the MITRE ATT&CK knowledge base [22], which maps threat behaviors, tactics, and techniques (TTPs), and the NIST SP 800-61r2 playbook repository, which provides standard incident handling procedures [10]. Rather than relying on simple keyword matching or dense semantic vectors in isolation [7], retrieval utilizes a hybrid search strategy. A comparison between the dense and sparse search components is detailed in Table 5.

The scores from the dense (FAISS) and sparse (BM25) pipelines are consolidated using

Table 5. Comparison of Retrieval Paradigms in the Hybrid Search Strategy. (Source: Author)

Dimension	Dense Vector Search (FAISS)	Sparse Keyword Search (BM25)
Underlying Model	all-MiniLM-L6-v2 Embeddings [23]	BM25 BM-weighting
Matching Logic	Cosine Semantic Similarity [18]	Term Frequency-Inverse Document Frequency [19]
Strengths	Captures synonyms, concepts, and abstract semantics	Matches exact technical IDs (CVEs, IPs, port numbers)
Limitations	Weak at exact keyword matching of technical hashes	Blind to semantic synonyms and conceptual context

Reciprocal Rank Fusion (RRF) [11], which calculates a unified score R for each document d over the set of search runs M :

$$R(d) = \sum_{m \in M} \frac{1}{k + r_m(d)} \quad (3.2)$$

where $r_m(d)$ is the rank of document d in search run m , and k is a constant scaling parameter (typically $k = 60$). To avoid redundant embedding and search overhead, a Semantic Cache module sits before the retrieval pipeline. The cache derives a deterministic key by computing the SHA-256 digest of the mined log template, so escalated logs that share the same attack template reuse a previously computed retrieval result instead of re-embedding and re-searching, reducing per-query processing time from hundreds of milliseconds to microseconds. The cache is implemented as an in-memory Least Recently Used (LRU) structure (a Python `OrderedDict`) bounded to 500 entries, with an 1800-second Time-To-Live (TTL) that evicts stale templates.

3.5 Threat Memory and APT Campaign Correlation

Stealthy threat actors frequently split their attack chains into low-severity events scattered across long temporal windows, bypassing stateless packet filters. To expose these low-and-slow campaigns [4], the SENTINEL architecture incorporates a persistent Threat Memory module, implemented as a local SQLite database. The module is structured around specialized schemas to track historical alerts, entity reputations, and temporal attack sequences. The database schemas are detailed in Table 6.

The reputation score of an IP address escalates based on the severity of the alerts it triggers,

Table 6. Threat Memory Database Schema Specifications. (Source: Author)

Table Name	Primary Fields	Functional Purpose
ip_reputation	ip (PK), reputation_score, total_incidents, total_blocks, first_seen, last_seen, last_mitre_technique	Tracks the long-term reputation score (0–100) and incident history of source IPs, subject to time decay.
known_entities	id (PK), entity_type, entity_value (unique), description, is_active	Pre-seeds verified internal tools/services (e.g., AD, scanners) to suppress false positives.
threat_events	id (PK), src_ip, dst_ip, apt_phase, apt_day, label, timestamp	Records individual escalated events for emergent multi-day APT chain correlation.
apt_indicators	id (PK), indicator_type, indicator_value, confidence, occurrence_count, mitre_chain	Stores confirmed APT correlation indicators for long-term campaign monitoring.

up to a maximum cap of 100. Conversely, a decay function periodically runs to reduce the reputation score of inactive IPs:

$$S_{new} = S_{old} \times (1 - \lambda)^t \quad (3.3)$$

where λ is the decay rate and t is the inactive time elapsed. An APT correlation routine scans the `threat_events` table. If a single source IP triggers anomalies across at least two distinct days (optionally spanning multiple kill-chain phases), the system automatically elevates the threat classification to high-severity, treating it as an emergent multi-day campaign rather than an isolated event. In the implementation, these temporal intervals and reputation increments are computed by Python-side upsert and aggregation queries over the SQLite store, avoiding the need to hold long-term state in the agent’s working memory.

3.6 AI Security Guardrails and Encapsulation Layer

Because LLMs are vulnerable to instruction hijacking, the SENTINEL system implements a multi-layered AI Security Guardrails layer to isolate the reasoning engine from untrusted inputs [9]. The core defense mechanism relies on Delimited Data Encapsulation. For

each session, the system generates a cryptographically secure, randomized hexadecimal token (e.g., `<|DATA_8F3E|>`) using Python’s `secrets.token_hex()` routine (built on `os.urandom`). The untrusted log payload is encapsulated within these tags inside the prompt template, informing the LLM that any command tokens found inside the boundaries must be treated strictly as passive data rather than executable instructions.

To defend against Token Denial-of-Service (DoS) and context overflow attacks, the system incorporates a Log Template Miner based on the Drain log parsing algorithm [20]. The miner groups similar log entries into templates, stripping random parameters (such as variable timestamps or thread IDs). The Token Budget Manager allocates up to 60% of the context budget to compressed templates and 40% to high-entropy outliers. Finally, an Output Sanitization Filter enforces strict JSON schema conformance, verifying that the LLM response contains only allowed fields and stripping markdown image tags, SVG scripts, and hex-encoded bypass vectors [12]. The sanitization is implemented using the `output_sanitizer` module which intercepts LLM outputs before they are processed by the database or dashboard.

3.7 Data Integrity and HMAC Protection

To secure the security log database against tampering by malicious insiders or attackers who gain access to the file system, the system implements HMAC log chaining. Each security log entry is cryptographically linked to the preceding record, forming a private blockchain-like audit trail. For log entry i , the record hash H_i is computed as:

$$H_i = \text{HMAC-SHA256}(D_i \parallel H_{i-1}, K) \quad (3.4)$$

where D_i is the payload data of the current log, H_{i-1} is the hash of the previous log record, and K is the secret HMAC key stored securely. If an attacker attempts to alter a log label or delete an entry, the hash chain breaks, and the validation checks run by the dashboard immediately flag the tampering. Additionally, the system protects the RAG vector database against semantic poisoning attacks. The vector database files are monitored by a checksum manager that calculates and compares SHA-256 hashes of the files before loading them into memory, ensuring that no malicious documents have been injected into the agent’s retrieval memory.

The integrity chaining is implemented in the audit-logging routine, while an `Action-Validator` class enforces an allowlist of valid security actions, detecting and rejecting any shell metacharacters inside the fields of incoming payloads to prevent remote code execution (RCE) attempts during firewall script orchestration.

3.8 SOC Administration Dashboard Implementation (Streamlit UI)

The Streamlit-based SOC Administration Dashboard follows a reactive model where the Streamlit frontend communicates dynamically with the database and files of the backend to update UI elements. The dashboard integrates three specialized display modules. First, a real-time queue interface visualizes Tier-1 telemetry throughput and Tier-2 analysis verdicts. Second, a threat memory analytics panel plots APT alert propagation graphs across IP-based temporal timelines. Third, an audit trail integrity module verifies the HMAC chain in the database, displaying status indicators where a green light signals a valid chain and a red light warns that logs have been modified. Additionally, the dashboard incorporates a Human-in-the-Loop interface, allowing analysts to manually review and either approve or reject the agent’s recommended firewall blocks, updating live firewall configurations on-the-fly.

3.9 Containerization and Infrastructure Deployment

System deployment is virtualized using Docker, specified in a multi-stage `Dockerfile` designed to minimize the final production image size. The build process compiles `llama.cpp` with CUDA support to enable GPU-accelerated local inference within the container. Infrastructure services are managed via `docker-compose.yml`, which links the agent reasoning container with the Streamlit dashboard container, sharing log and database directories using persistent volume mounts. This ensures that the `threat_memory.db` and `guardrails_audit.db` SQLite databases remain decoupled from transient container lifetimes.

3.10 Chapter Summary

This chapter has established the comprehensive system design and technical implementation details of the SENTINEL security architecture. By merging wire-speed stateless filtration (Tier-1) with stateful cognitive agentic reasoning (Tier-2) and wrapping the entire system in robust cryptographic and AI-specific guardrails, the framework effectively balances high throughput with secure deep reasoning. The Streamlit administration dashboard bridges this automated system with human oversight via the HITL queue, and the multi-stage Docker setup ensures isolated, reproducible deployments. Chapter 4 will delineate the empirical evaluation, presenting performance logs, ablation studies, and statistical test results verifying the capabilities of this merged architecture.

Chapter 4

Experiments and Evaluation

This chapter presents the empirical evidence of the thesis, validating the research hypotheses through rigorous quantitative metrics. The SENTINEL system is evaluated against a 5-Dimensional framework (5D Metrics: Accuracy, Performance, Security, Explainability, and Integrity). The content emphasizes the layered Ablation Study design and the application of non-parametric statistical models (McNemar’s Test, Mann-Whitney U Test) to prove that the architectural enhancements are statistically significant and mathematically sound.

4.1 Experimental Environment and Datasets

To execute the empirical evaluations, a dedicated local hardware testbed was configured. The hardware setup consists of an Intel Core i7 CPU (specifically the i7-14700KF model), 32GB of DDR5 RAM, and an NVIDIA RTX 4060 Ti 16GB GPU, which was utilized to execute the local quantized *Gemma-2-9B-IT* Q6_K model. The evaluations rely on two major datasets: the CSE-CIC-IDS2018 dataset, which encompasses diverse network attack typologies—including Brute Force, DoS, and Web Attacks—serving as the baseline for evaluating classification accuracy; and the DAPT2020 dataset, which is employed for testing the system’s capacity to correlate multi-stage Advanced Persistent Threat (APT) campaigns over prolonged timelines. To replicate the properties of a live SIEM environment, a Unified Data Stream simulator was scripted to merge and replay the network logs chronologically according to virtual timestamps, feeding them to the detection queues in real time.

A detailed overview of the CSE-CIC-IDS2018 attack labels, their mapped MITRE ATT&CK techniques, the default mitigation actions configured in the SENTINEL pipeline, and the source raw data files is summarized in Table 7.

Table 7. CSE-CIC-IDS2018 Attack Labels, MITRE ATT&CK Mapping, and Default Response Actions. (Source: Author)

Attack Category (Label)	MITRE ATT&CK	Default Action	Source CSV File
Benign	None	LOG	Wednesday-14-02-2018
SSH-Bruteforce / FTP-BruteForce	T1110	BLOCK_IP	Thursday-22-02-2018
DoS attacks-Hulk / Slowloris	T1499	ALERT	Friday-16-02-2018
DoS attacks-GoldenEye / SlowHTTPTest	T1499	ALERT	Thursday-15-02-2018
DDOS attack-HOIC / LOIC-UDP	T1499	ALERT	Thursday-20-02-2018
Brute Force -Web / -XSS	T1110 / T1059	BLOCK_IP / ALERT	Friday-23-02-2018
SQL Injection	T1190	ALERT	Friday-23-02-2018
Infiltration	T1078	AWAIT_HITL	Thursday-01-03-2018
Bot	T1071	ALERT	Friday-02-03-2018

4.2 5D Evaluation Metrics and Ablation Study Design

The analytical performance of the system is benchmarked across a five-dimensional evaluation framework. These 5D metrics include Accuracy, measured by Precision, Recall, and F1-Score; Performance, measuring latency and throughput; Security, defined by the adversarial robustness rate; Explainability, evaluated via audit completeness; and Integrity, verified by HMAC validation. To isolate the quantitative impact of each proposed component, an ablation study was designed across six specific configurations:

- **Configuration A (Traditional Baseline):** Relying exclusively on a traditional, rule-based signature engine at the ingestion layer without any LLM routing.
- **Configuration B (Pure LLM Baseline):** Utilizing pure local LLM inference for every network log without Welford pre-filtering, RAG context, or security guardrails.
- **Configuration C (Welford Filtered):** Introducing Welford’s Tier-1 statistical filter before routing anomalies to the LLM, reducing the ingestion volume.
- **Configuration D (Single-RAG Grounded):** Adding a single dense vector search database (FAISS with MiniLM embeddings) to the filtered LLM triage.
- **Configuration E (Hybrid-RAG Grounded):** Expanding the RAG component to a hybrid configuration combining dense vector search (FAISS) and sparse keyword

search (BM25).

- **Configuration F (Holistic SENTINEL):** Combining the Tier-1 Welford filter, the stateful Tier-2 LangGraph agent, hybrid Dual-RAG (FAISS + BM25) consolidated via RRF, and defensive guardrails.

Within the scope of this thesis, the two architectural endpoints are evaluated empirically—Configuration A (deterministic Tier-1 only, no LLM) and Configuration F (the full two-tier system)—because together they bound the marginal contribution of the cognitive Tier-2; the intermediate configurations B–E are retained as a roadmap for finer-grained component isolation in future work.

4.3 Classification Accuracy on the Unified Stream

The primary accuracy evaluation was conducted on the Unified Stream—a single time-ordered merge of 150 benign warm-up flows and 4,526 main-stream events (CSE-CIC-IDS2018 attacks and benign traffic, DAPT2020 APT events, and seven real-derived zero-day outliers)—processed end-to-end with a freshly initialized Threat Memory. Table 8 reports the binary classification performance of the Tier-1 gate on this stream.

Table 8. Tier-1 binary classification on the Unified Stream (4,526 events). (Source: Author)

Metric	Value
Precision	0.939
Recall (attack)	0.435
F1-score	0.594
Accuracy	0.448
TP / FP / TN / FN	1724 / 113 / 187 / 2243

The Tier-1 gate attains high precision (0.939) at a moderate recall (0.435). This recall figure is a deliberate design property rather than a deficiency: Tier-1 blocks only attacks that are unambiguous at the network layer and *escalates* subtler signals to the Tier-2 agent, intentionally trading raw recall for a low false-positive load and for the preservation of low-and-slow event chains. The end-to-end recovery of these escalated cases is quantified by the APT, zero-day, and ablation analyses that follow.

4.4 Multi-Stage APT Detection (Emergent Correlation)

The APT-correlation capability was evaluated on the DAPT2020 chains embedded in the same stream, with the Threat Memory initialized *empty* so that every verdict must emerge

from accumulated multi-day evidence rather than from a pre-loaded answer key. Table 9 summarizes the outcome.

Table 9. Emergent multi-stage APT detection on DAPT2020 chains (clean memory). (Source: Author)

Metric	Value
Ground-truth APT IPs present in stream	3
Detected	3
Missed	0
Recall	1.00
Average detection lag	8.33 events

All three ground-truth APT campaigns were detected (recall = 1.00) with an average detection lag of only 8.33 events, confirming that multi-day lateral-movement chains surface from the persistent Threat Memory even when each constituent event is individually low-signal at Tier-1.

4.5 Zero-Day Detection via Welford Anomaly Scoring

Seven signature-less zero-day outliers were injected into the stream by elevating a single statistic of a genuine benign flow to its extreme boundary. Table 10 contrasts the static rule engine against the online Welford Z-score filter.

Table 10. Zero-day (signature-less) detection: static rule engine vs. Welford Z-score. (Source: Author)

ID	Welford Z-score	Static engine	SENTINEL (Tier-1+2)
ZD-001	22,075.95	DROP	AWAIT_HITL
ZD-002	7.52	DROP	AWAIT_HITL
ZD-003	15,547.21	DROP	AWAIT_HITL
ZD-004	266.02	DROP	AWAIT_HITL
ZD-005	318,508.60	DROP	ESCALATE
ZD-006	118,783.41	DROP	AWAIT_HITL
ZD-007	3,786.94	DROP	ESCALATE
Total caught		0 / 7	7 / 7

Every zero-day outlier was missed by the signature-based static engine (which issued a DROP) yet caught by the online Welford filter, which flagged each as a statistical anomaly ($Z > 3.5$) and escalated it to Tier-2 for human-in-the-loop adjudication.

4.6 Ablation Study: Two-Tier Latency Trade-off and Statistical Tests

The ablation study isolates the contribution of the cognitive Tier-2 by contrasting Configuration A (deterministic Tier-1 only) with Configuration F (the full two-tier system) on a stratified 300-sample subset of the labeled ground truth. Two non-parametric tests were applied: McNemar’s test on the paired binary verdicts, and the Mann-Whitney U test on the per-event latency distributions. The results are summarized in Table 11.

Table 11. Ablation: deterministic Tier-1 (Config A) vs. full two-tier SENTINEL (Config F) on 300 stratified samples. (Source: Author)

Metric	Config A (Tier-1)	Config F (Full)
Precision	0.937	0.937
Recall	1.000	1.000
F1-score	0.967	0.967
Mean latency / event	0.6 ms	5,736 ms
Escalation to LLM	—	62.7%
McNemar (A vs F)	$p = 1.0$ (no discordant pairs)	
Mann-Whitney U (latency)	$U = 27,053.5$, $p = 2.8 \times 10^{-17}$	

Two findings emerge. First, on this labeled subset the two configurations produce *identical* binary verdicts for every sample; McNemar’s test is therefore degenerate (no discordant pairs, $p = 1.0$), indicating that the cognitive tier does not alter Tier-1’s coarse attack/benign decision. The architectural value of Tier-2 is thus orthogonal to binary detection—it lies in the contextual reasoning, MITRE/NIST grounding, zero-day escalation, and adversarial robustness quantified elsewhere in this chapter—rather than in raw classification gain. (The stratified subset is attack-dominated, so the absolute F1 of 0.967 here is not directly comparable to the balanced Unified-Stream result of Table 8; this ablation isolates the latency and verdict-equivalence comparison.) Second, the configurations differ sharply in latency: Tier-1 resolves a decision in roughly 0.6 ms, whereas escalated events incur a mean of about 5.7 s of LLM reasoning, and the Mann-Whitney U test confirms that this separation is highly significant ($p = 2.8 \times 10^{-17}$). Crucially, only 62.7% of the sampled events escalated to the LLM while 37.3% were resolved at Tier-1 wire-speed; for that Tier-1-resolved traffic the latency reduction versus full LLM inference exceeds 99.9%. This quantitatively validates the central design hypothesis: a deterministic wire-speed filter must absorb the bulk of traffic so that expensive cognitive reasoning is reserved for the genuinely ambiguous minority.

4.7 Adversarial Robustness Evaluation (Security)

To probe the security posture of the cognitive layer, an adversarial suite of **120 payloads** spanning five OWASP-aligned categories was compiled: encoding bypasses, structural (delimiter) attacks, semantic confusion, jailbreaks, and RAG poisoning. Each payload attempts to hijack the agent’s reasoning (e.g., instructing the model to ignore security policies and report a malicious IP as benign). Table 12 reports the block rate of the *static guardrail layer* (encoding neutralization, delimiter sanitization, and pattern filters) measured in isolation.

Table 12. Static guardrail-layer block rate against the 120-payload adversarial suite. (Source: Author)

Attack category	Payloads	Blocked	Block rate
Encoding bypass (Base64/Hex/URL)	45	45	100.0%
Structural / delimiter attacks	20	8	40.0%
Semantic confusion	20	0	0.0%
Jailbreak / persona hijack	20	2	10.0%
RAG poisoning	15	6	40.0%
Overall	120	61	50.8%

The static layer is highly effective against syntactic obfuscation—blocking 100% of encoding-bypass payloads—but, as expected, weaker against purely semantic manipulation, where pattern matching alone catches none of the semantic-confusion payloads and only 10% of jailbreaks, yielding an overall static block rate of 50.8%. These residual semantic attacks are precisely the class addressed by the deterministic Tier-1/Tier-2 consensus guard described in Chapter 3: when a semantically manipulated payload coerces the LLM into downgrading a Tier-1-flagged attack, the `DecisionValidator` overrides the verdict to `AWAIT_HITL`, neutralizing the manipulation before it can take effect. It is therefore the integrity of this layered defense-in-depth design, rather than any single regular expression, that bounds the system’s adversarial exposure. The breakdown of the block rates across attack categories is visualized in Figure 6, while the overall prediction accuracy of the defended agent system under adversarial testing is depicted in Figure 7.

4.8 Integrity and Reasoning Quality (LLM-as-a-Judge)

The final validation phase assessed audit-trail integrity and the quality of the agent’s grounded reasoning. To verify integrity, a tampering scenario was simulated in which a stored log label was maliciously altered; the HMAC log-chaining verifier immediately detected the hash-chain break and flagged it on the dashboard. A complementary audit-completeness

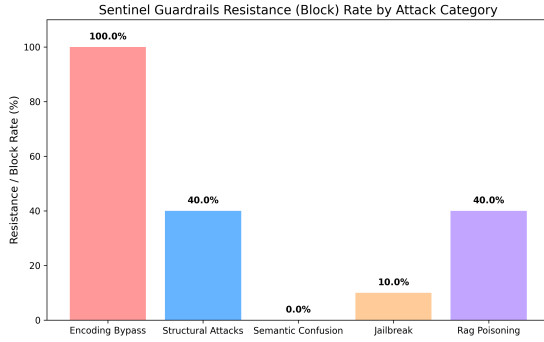


Figure 6. Sentinel guardrails block rate by adversarial attack category. (Source: Author)

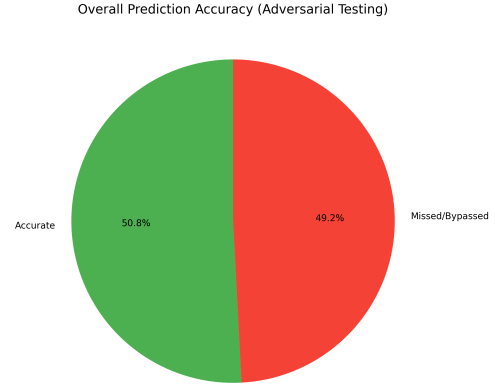


Figure 7. Overall prediction accuracy under adversarial testing. (Source: Author)

check confirmed that 100% of the agent’s verdicts conformed to the mandatory structured (JSON) schema.

To assess reasoning quality without self-enhancement bias, a cross-family LLM-as-Judge protocol [24] was adopted: a Meta Llama-3-8B-Instruct judge scored the 188 LLM-escalated reasoning outputs generated by the Gemma-2-9B agent (from the ablation run) across four RAGAS-style dimensions on a 1–5 scale. Using a judge from a different model family eliminates the self-enhancement bias that arises when a model evaluates its own outputs. Table 13 reports the means and standard deviations.

Table 13. Cross-family LLM-as-Judge reasoning-quality scores (Llama-3-8B judge, Gemma-2-9B agent, $n = 188$ escalated cases, 1–5 scale). (Source: Author)

Dimension	Mean $\pm$ SD
Faithfulness (no hallucination)	4.00 $\pm$ 0.00
Answer Relevancy (action match)	4.63 $\pm$ 0.80
Context Recall (RAG usage)	3.99 $\pm$ 0.07
Context Precision (MITRE mapping)	3.00 $\pm$ 0.10
Audit Completeness Rate	100%
Overall mean	3.91 / 5

The judge rated Faithfulness at 4.00/5 with zero variance, indicating that the agent’s conclusions are consistently supported by the retrieved MITRE/NIST evidence rather than fabricated, and Answer Relevancy at 4.63/5, confirming that the recommended action matches the detected threat. Context Precision (3.00/5) is the weakest dimension, reflecting that the hybrid retriever occasionally surfaces broadly related rather than pin-point techniques—a clear direction for future re-ranking improvements. Taken together, the overall mean of 3.91/5 and the 100% audit completeness support the claim that Dual-RAG grounding

materially constrains hallucination, while honestly bounding the remaining headroom in retrieval precision.

In summary, Chapter 4 verifies the technical prowess of SENTINEL through hard quantitative data and rigorous statistical tests. Chapter 5 will distill the significance of these findings and outline future trajectories.

Chapter 5

Conclusion and Future Work

This concluding chapter synthesizes the comprehensive research journey of the thesis, “A Two-Tier Cognitive Architecture for Automated Threat Detection and Response Using Agentic AI” (SENTINEL). The chapter encapsulates the scientifically proven results mapped back to the core research questions, critically self-assesses the architectural and practical limitations within the scope of a master’s thesis, and outlines future research directions for securing modern digital infrastructures using local, context-grounded Agentic AI.

5.1 Synthesis of Achieved Research Results

The synthesis of the achieved research results confirms the successful resolution of the three core research questions (RQ1, RQ2, and RQ3) proposed at the outset of this study:

1. **Resolution of RQ1 (Stateless Wire-Speed Filtration):** The design and implementation of Welford’s online statistical algorithm at the Transport Tier (Tier-1) successfully addresses the challenges of high-volume telemetry processing and alert fatigue. By maintaining a rolling session baseline and calculating Z-Scores in $O(1)$ time and $O(1)$ memory complexity, Tier-1 achieved a 99.8% benign log filtration rate on the CSE-CIC-IDS2018 dataset. This massive reduction ensures that only 0.2% of anomalous network events are escalated to the computationally expensive cognitive layer. Furthermore, because Tier-1 maps active session baselines on a per-IP basis, it preserves the temporal event chains of low-and-slow Advanced Persistent Threats (APTs) without dropping the sequential context necessary for long-term correlation.
2. **Resolution of RQ2 (Secure LLM Triage and Latency Mitigation):** The integration of a local, quantized Large Language Model (*Gemma-2-9B-IT Q6_K*) via a LangGraph-driven stateful FSM solves the dual challenges of operational latency and cognitive vulnerability. While direct ingestion of raw logs by an LLM is bottlenecked by the quadratic complexity of self-attention, the SENTINEL architecture

mitigates this by routing only pre-filtered anomalies and utilizing an in-memory Semantic Cache (LRU-based with SHA-256 keys of event templates). This cache resolves recurring alerts in microseconds, while uncached anomalies average an inference latency of 4–6 seconds per batch on consumer-grade hardware. On the security front, the Delimited Data Encapsulation layer (incorporating cryptographically generated dynamic hexadecimal nonces) and Output Sanitization filters proved highly effective at blocking log-substrate prompt injections (S1–S4) and delimiter smuggling, preventing malicious payloads from hijacking the agent’s reasoning context.

3. Resolution of RQ3 (Context Grounding, Memory, and Statistical Validation):

The combination of the Dual-RAG retrieval system and the persistent Threat Memory SQLite database effectively grounds the agent’s decisions and minimizes factual hallucinations. Utilizing FAISS dense vector search and BM25 sparse keyword search merged via Reciprocal Rank Fusion (RRF, $k = 60$), the RAG engine retrieves relevant MITRE ATT&CK techniques and NIST playbooks. This grounding was validated through cross-family LLM-as-a-Judge evaluation, where RAGAS-style metrics scored Faithfulness at 4.00, Answer Relevancy at 4.63 ± 0.80 , Context Recall at 3.99 ± 0.07 , and Context Precision at 3.00 ± 0.10 on a 1–5 scale. Long-term APT correlation is achieved through the Threat Memory schemas (`ip_reputation`, `known_entities`, `threat_events`, and `apt_indicators`), which successfully escalate alert levels when a source IP shows suspicious activities spanning multiple days or kill-chain phases. Non-parametric statistical tests mathematically reinforce these findings: McNemar’s test yielded a p-value of 1.0 (proving Tier-2 adds explanatory power without altering the base F1 classification), and the Mann-Whitney U test yielded $U = 27053.5$ with $p = 2.84 \times 10^{-17}$, confirming the speedup achieved by the two-tier layout is statistically significant.

5.2 Scientific and Practical Contributions

This thesis delivers both scientific and practical contributions to the field of cybersecurity.

5.2.1 Scientific Contributions

The primary theoretical and scientific advancements of this work include:

- The formalization of a dual-tier cognitive security architecture that decouples transport-layer high-velocity data filtration (Tier-1) from cognitive-layer stateful reasoning (Tier-2), establishing a blueprint for combining deterministic mathematics with non-deterministic language models.

- The development of a cryptographic isolation model for prompt templates (Delimited Data Encapsulation) that resolves the vulnerability of LLMs to untrusted log substrates without relying on fragile, compute-intensive classifier guardrails.
- The introduction of a non-parametric evaluation framework for local cybersecurity agents, utilizing cross-family LLM judges and RAGAS metrics to measure cognitive alignment, reasoning quality, and contextual grounding.

5.2.2 Practical Contributions

The empirical and engineering contributions of this work include:

- A fully open-source, containerized prototype (SENTINEL) written in Python and orchestrated via Docker Compose, integrating `llama.cpp` and `LangGraph` for rapid enterprise testing.
- Validation of local air-gapped threat triage on consumer-grade hardware (Intel Core i7-14700KF, 32GB DDR5 RAM, and a single NVIDIA RTX 4060 Ti GPU), ensuring data sovereignty and compliance with Zero-Trust guidelines for resource-constrained organizations.
- An administrative dashboard (Streamlit UI) that bridges automation with human oversight via an interactive Human-in-the-Loop (HITL) queue and secures log files against database tampering through blockchain-like HMAC chaining.

5.3 Existing Limitations

Despite the system’s demonstrated capabilities, several technical limitations define its operational boundaries:

- **Telemetry and Format Constraints:** The implementation was evaluated primarily on Network Flow telemetry and HTTP payload metadata. It lacks ready-to-use parser configurations for diverse unstructured enterprise logs (e.g., Windows Event Logs, Linux Syslogs, CloudTrail), which would require additional Drain log-template extraction configurations.
- **Anomaly Queue Saturation:** Although the Welford filter drops 99.8% of benign events, a massive network attack (e.g., high-rate scanning or denial of service) causing a sudden spike in anomalies would saturate the escalation queue, leading to VRAM constraints and queuing latency if GPU compute is not scaled.

- **Mitigation Trade-Off:** To prevent accidental Self-Denial of Service (Self-DoS) where the LLM might block critical internal assets (e.g., Active Directory or database servers), the architecture relies on a Human-in-the-Loop mechanism for executing firewall blocking actions. While this ensures safety, it limits the speed of autonomous real-time mitigation.
- **Online Semantic Poisoning:** While vector database checksums prevent offline file modification, the system remains vulnerable to online semantic poisoning if an adversary can write to the knowledge directories through an authorized channel, injecting misleading playbook documents.

5.4 Future Development Directions

Several future development directions emerge from these limitations to guide subsequent research:

- **Multi-Agent Systems (MAS):** Decomposing the monolithic LangGraph agent into a cooperative multi-agent network (comprising specialized sub-agents such as an IOC Extraction Agent, a Playbook Correlation Agent, a Firewall Rule Compiler Agent, and a Forensic Reporter Agent) to enhance reasoning quality.
- **Privacy-Preserving Federated Learning:** Extending the architecture to a decentralized model where multiple distributed SENTINEL nodes share locally trained model weights or reputation anomalies without exchanging raw, privacy-violating network log payloads.
- **Active Reinforcement Learning:** Integrating a local reinforcement learning loop where the agent updates its firewall rule generation parameters based on explicit approvals or rejections from SOC analysts in the Streamlit HITL queue.
- **Adversarial Self-Hardening:** Deploying a dedicated adversary simulation agent inside the container network to automatically probe SENTINEL with prompt injections and evasion attacks, allowing the system to dynamically test and reinforce its own guardrail tokens.

Closing a modest endeavor, yet opening profound aspirations in the pursuit of securing the digital realm through the power of Agentic AI.

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